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## Inclusion of time-series geospatial markers in analyses employing a cyber-decision platform


### Abstract

A system for programmatic and user interactive inclusion of time-series geospatial markers in analyses employing a cyber-decision platform has been developed comprising a module to retrieve an indexed geospatial image tiles and map overlay data corresponding to tiles. The system may attach a unique geo-hash label to each point of an indexed geospatial image tile based upon the point's geographical coordinates such that points in close proximity will have similar geo-hash values. A web application interface allows users to interactively retrieve and visualize indexed geospatial image tiles, add map overlays and create geo-hashes as desired.



**Inventors:** **Crabtree; Jason (Vienna, VA), Sellers; Andrew (Monument, CO)**

**Applicant:** **QOMPLX LLC** (Reston, VA)

**Family ID:** **73650417**

**Assignee:** **QOMPLX LLC (Reston, VA)**

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*Primary Examiner:* Mackes; Kris E

*Attorney, Agent or Firm:* Galvin Patent Law LLC

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) Priority is claimed in the application data sheet to the following patents or patent applications, each of which is expressly incorporated herein by reference in its entirety: Ser. No. 17/106,997 Ser. No. 15/931,534 Ser. No. 16/777,270 Ser. No. 16/720,383 Ser. No. 15/823,363 Ser. No. 15/725,274 Ser. No. 15/655,113 Ser. No. 15/616,427 Ser. No. 14/925,974 Ser. No. 15/237,625 Ser. No. 15/206,195 Ser. No. 15/186,453 Ser. No. 15/166,158 Ser. No. 15/141,752 Ser. No. 15/091,563 Ser. No. 14/986,536 Ser. No. 15/683,765 Ser. No. 15/409,510 Ser. No. 15/379,899 Ser. No. 15/376,657 Ser. No. 16/718,906 Ser. No. 15/879,182 Ser. No. 15/850,037 Ser. No. 15/673,368 Ser. No. 15/489,716 Ser. No. 15/905,041 Ser. No. 16/191,054 Ser. No. 16/654,309 Ser. No. 15/847,443 Ser. No. 15/790,457 62/568,298 Ser. No. 15/790,327 62/568,291 Ser. No. 16/660,727 Ser. No. 15/229,476

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention is in the field of use of computer systems in business information management, operations and predictive planning Specifically, including geospatial data or time related geospatial data during the use of a cyber decision system for business operations.

Discussion of the State of the Art

(2) Nothing has augmented our ability to gather and store information analogous to the rise of electronic and computer technology. There are sensors of all types to measure just about any condition one may imagine. Computers have allowed the health information for a large portion of the human population to be stored and accessible. Similarly, mass quantities of detailed data concerning the working of government, economics, demographics, climate change and population shifts are all being continuously stored for public analysis and is available on the world wide web. Further examples abound, but the point has been made. The meteoric rise of computer networking and the internet has only served to turn the accrual of information into a torrent as now huge populations may exchange observations, data and ideas, are even invited to do so; vast arrays of sensors may be tied together in meaningful ways all generating data which may be stored for future analysis and use. The receipt and storage of data has gotten to the point where an expert has been quoted as estimating that as much data is currently accrued in two days as was accrued in all history prior to 2003 (Eric Schmidt, Google). While not all of this information would benefit from

the ability to either tag the geographical location from which is originated or to add geospatial imagery of the location or region of origin to it, a large proportion would be augmented by such an addition. The addition of either a geographical tag or image specific for its location of origin opens up the possibility of later retrieval of all data, regardless of time collected or event type on the basis of location for inclusion in predictive analytics, allowing the possibility of entirely new and significant outcomes and conclusions to be discovered where such correlations would not have been possible, solely due to geographical tagging or the inclusion of location specific, possibly map overlaid geographical image tiles.

(3) What is needed is a fully integrated system that allows the assignment of location or region specific geographical tags, indexed geospatial image tiles and corresponding map overlay data to information of a plurality of types and subject matter from many heterogeneous sources using a scalable, expressively scriptable, connection interface, identifies and analyzes that high volume data, transforming it into a useful format. Such a system may then use that data, the geographical data serving as one characteristic to drive an integrated, highly scalable simulation engine which may employ combinations of the system dynamics, discrete event and agent based paradigms within a simulation run such that the most useful and accurate data transformations are obtained and stored for the human analysts to rapidly digest the presented information, readily comprehend any predictions or recommendations and then creatively respond to optimize the business decision process including security, expansion, growth and profit. This multimethod information capture, analysis, transformation, outcome prediction, and presentation system constitute in effect a "business operating system."

SUMMARY OF THE INVENTION

(4) Accordingly, the inventor has conceived and reduced to practice a system for inclusion of time-series geospatial markers in analyses employing a cyber-decision platform. The following non-limiting summary of the invention is provided for clarity, and should be construed consistently with embodiments described in the detailed description below.

(5) In a typical embodiment, the business operating system, with the inclusion of an indexed global tile module and geospatial tile manager allows both newly accruing data and previously archived data to be during simplest usage tagged with a unique geo-hash based upon the geographic coordinates of the data's origin or as determined, an indexed geospatial image tile to be attached to the data. Further, map overlays corresponding to the attached indexed geospatial tiles may also be included. Addition of geographical references to both current accruing data and previously archived data then allows inclusion of data from varying times and events previously thought to be unrelated to be predictively analyzed and visualized based on a common location, possibly leading to new highly significant insights and business solutions. Also possible are time-series analyses of geospatial image tiles either electronically or by human inspection such as expert analyst or crowdsourcing of those time related images for specific markers or changes related to a specific analysis. The addition of geographical data is expected to augment not only strictly real-world predictive analysis but also simulation analysis of specific business decisions which may then be used outright or may be compared, simulated progression to real-world observed outcome to measure model assumption accuracy or hard to discern progress of an ongoing business project.

(6) According to a preferred embodiment of the invention, a system for inclusion of time-series geospatial markers in analyses employing a cyber-decision platform has been devised and reduced to practice comprising: an indexed geospatial tile module stored in a memory of and operating on a processor of a computing device and configured to retrieve a plurality of indexed geospatial image tiles from a plurality of sources, retrieve a plurality of map overlay data corresponding to at least a portion of the retrieved indexed geospatial tile data from a plurality of sources, calculate a unique geo-hash label for each point on each indexed geospatial tiles based upon the point's geographical coordinates and such that points in close proximity in geographical location will have greater similarity in geo-hash designation than those more distant from each other, is used to attach geo-

hash data or full indexed geospatial tile data or both to at least a portion of available archival raw event data, at least a portion of archival predictive analyses result data, and at least a portion of event data being obtained in real time as part of a predictive analysis campaign, carried out by cyber-decision platform; and a web application-based interface module stored in a memory of and operating on a processor of a computing device and configured to allow interactive users to retrieve and visualize indexed geospatial image tiles for desired geographic unit from a plurality of sources, allow interactive users to retrieve map overlay data for at least one of the retrieved indexed geospatial image tiles, allow the interactive user to calculate geo-hashes for and geographical points, mediate attachment of geospatial data to existing data for inclusion as part of a predictive analysis performed by cyber-decision platform.
(7) According to a preferred embodiment of the invention, a system for a system for inclusion of time-series geospatial markers in analyses employing a cyber-decision platform has been devised and reduced to practice. Wherein the geographic unit is a geographical location made up of at least one geographical point. Wherein the geographic unit is a geographical region made up of a plurality of geographical locations. Wherein at least of portion of the indexed geospatial tile data represent time-series image tile collections corresponding to the time progression of the predictive analyses carried out by cyber-decision platform. Wherein at least one of the indexed geospatial image tiles is overlaid with the corresponding map overlay data. Wherein at least one of the indexed geospatial image tiles retrieved by user of web application-based interface module is presented directly through electronic or other, possibly paper-based, output. Wherein at least one of the indexed geospatial image tiles retrieved by user of web application-based interface module is overlaid with corresponding map overlay data and presented directly through electronic or other, possibly paper-based, output. Wherein at least one set of the indexed geospatial image tiles for the same geographical location at progressive time points are retrieved by user of web application-based interface module are presented, at least a portion also displaying corresponding map overlay data directly through electronic or other, possibly paper-based, output. Wherein at least a portion of the retrieved geospatial image tile data is used solely to calculate at least one geo-hash to be attached to data prior to storage to allow later geo-hash specific manipulation of so labelled data.
(8) According to a preferred embodiment of the invention, method for inclusion of time-series geospatial markers in analyses employing a cyber-decision platform comprising the steps of: a) retrieving indexed geospatial image tiles from a plurality of sources using an indexed global tile module stored in a memory of and operating on a processor of a computing device; b) retrieving map overlay data corresponding to predetermined indexed geospatial image tiles from a plurality of sources using an indexed global tile module stored in a memory of and operating on a processor of a computing device; c) employing, programmatically, at least a portion of the retrieved indexed geospatial image tiles and corresponding map overlay data; d) employing, interactively, at least a portion of the retrieved indexed geospatial image tiles and corresponding map overlay data using a web application-based interface module stored in a memory of and operating on a processor of a computing device.

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## Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES
(1) The accompanying drawings illustrate several embodiments of the invention and, together with the description, serve to explain the principles of the invention according to the embodiments. One skilled in the art will recognize that the particular embodiments illustrated in the drawings are merely exemplary, and are not intended to limit the scope of the present invention.
(2) FIG. **1** is a diagram of an exemplary architecture of a business operating system an indexed global tile module according to an embodiment of the invention.

(3) FIG. **2** is a diagram of the web application-based interface for an indexed global tile module as per one embodiment of the invention.
(4) FIG. **3** is a diagram of an indexed global tile module as per one embodiment of the invention
(5) FIG. **4** is a flow diagram illustrating the function of the indexed global tile module as per one embodiment of the invention
(6) FIG. **5** is a block diagram illustrating an exemplary hardware architecture of a computing device used in various embodiments of the invention.
(7) FIG. **6** is a block diagram illustrating an exemplary logical architecture for a client device, according to various embodiments of the invention.
(8) FIG. **7** is a block diagram illustrating an exemplary architectural arrangement of clients, servers, and external services, according to various embodiments of the invention.
(9) FIG. **8** is another block diagram illustrating an exemplary hardware architecture of a computing device used in various embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(10) The inventor has conceived, and reduced to practice a system for the inclusion of time-series geospatial markers in analyses employing a cyber-decision platform.
(11) One or more different inventions may be described in the present application. Further, for one or more of the inventions described herein, numerous alternative embodiments may be described; it should be understood that these are presented for illustrative purposes only. The described embodiments are not intended to be limiting in any sense. One or more of the inventions may be widely applicable to numerous embodiments, as is readily apparent from the disclosure. In general, embodiments are described in sufficient detail to enable those skilled in the art to practice one or more of the inventions, and it is to be understood that other embodiments may be utilized and that structural, logical, software, electrical and other changes may be made without departing from the scope of the particular inventions. Accordingly, those skilled in the art will recognize that one or more of the inventions may be practiced with various modifications and alterations. Particular features of one or more of the inventions may be described with reference to one or more particular embodiments or figures that form a part of the present disclosure, and in which are shown, by way of illustration, specific embodiments of one or more of the inventions. It should be understood, however, that such features are not limited to usage in the one or more particular embodiments or figures with reference to which they are described. The present disclosure is neither a literal description of all embodiments of one or more of the inventions nor a listing of features of one or more of the inventions that must be present in all embodiments.
(12) Headings of sections provided in this patent application and the title of this patent application are for convenience only, and are not to be taken as limiting the disclosure in any way.
(13) Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more intermediaries, logical or physical.
(14) A description of an embodiment with several components in communication with each other does not imply that all such components are required. To the contrary, a variety of optional components may be described to illustrate a wide variety of possible embodiments of one or more of the inventions and in order to more fully illustrate one or more aspects of the inventions. Similarly, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may generally be configured to work in alternate orders, unless specifically stated to the contrary. In other words, any sequence or order of steps that may be described in this patent application does not, in and of itself, indicate a requirement that the steps be performed in that order. The steps of described processes may be performed in any order practical. Further, some steps may be performed simultaneously despite being described or implied as occurring sequentially (e.g., because one step is described after the

other step). Moreover, the illustration of a process by its depiction in a drawing does not imply that the illustrated process is exclusive of other variations and modifications thereto, does not imply that the illustrated process or any of its steps are necessary to one or more of the invention(s), and does not imply that the illustrated process is preferred. Also, steps are generally described once per embodiment, but this does not mean they must occur once, or that they may only occur once each time a process, method, or algorithm is carried out or executed. Some steps may be omitted in some embodiments or some occurrences, or some steps may be executed more than once in a given embodiment or occurrence.

(15) When a single device or article is described, it will be readily apparent that more than one device or article may be used in place of a single device or article. Similarly, where more than one device or article is described, it will be readily apparent that a single device or article may be used in place of the more than one device or article.

(16) The functionality or the features of a device may be alternatively embodied by one or more other devices that are not explicitly described as having such functionality or features. Thus, other embodiments of one or more of the inventions need not include the device itself.

(17) Techniques and mechanisms described or referenced herein will sometimes be described in singular form for clarity. However, it should be noted that particular embodiments include multiple iterations of a technique or multiple manifestations of a mechanism unless noted otherwise. Process descriptions or blocks in figures should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of embodiments of the present invention in which, for example, functions may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those having ordinary skill in the art.

(18) Program functions and capabilities are not always attributed to a named software set or library. This in no instance implies that such a specific program, program function, or code library is not employed but is meant to allow time progression-based changes to be made. In all cases at least one open source or proprietary software package providing the attributed functional result may be available and known to those skilled in the art or the algorithm needed to accomplish the function determinable by those skilled in the art.

(19) Conceptual Architecture

(20) FIG. **1** is a diagram of an exemplary architecture of a business operating system including an indexed global tile module **100** according to an embodiment of the invention. Client access to the system **105** for specific data entry, system control and for interaction with system output such as automated predictive decision making and planning and alternate pathway simulations, occurs through the system's distributed, extensible high bandwidth cloud interface **110** which uses a versatile, robust web application driven interface for both input and display of client-facing information and a data store **112** such as, but not limited to MONGODB™, COUCHDB™, CASSANDRA™ or REDIS™ depending on the embodiment. Much of the business data analyzed by the system both from sources within the confines of the client business, and from cloud based sources **107**, public or proprietary such as, but not limited to: subscribed business field specific data services, external remote sensors, subscribed satellite image and data feeds and web sites of interest to business operations both general and field specific, also enter the system through the cloud interface **110**, data being passed to the connector module **135** which may possess the API routines **135***a* needed to accept and convert the external data and then pass the normalized information to other analysis and transformation components of the system, the directed computational graph module **155**, high volume web crawler module **115**, multidimensional time-series database **120** and the graph stack service **145**. The directed computational graph module **155** retrieves one or more streams of data from a plurality of sources, which includes, but is not limited to, a plurality of

physical sensors, network service providers, web based questionnaires and surveys, monitoring of electronic infrastructure, crowd sourcing campaigns, and human input device information. Within the directed computational graph module **155**, data may be split into two identical streams in a specialized pre-programmed data pipeline **155***a*, wherein one sub-stream may be sent for batch processing and storage while the other sub-stream may be reformatted for transformation pipeline analysis. The data is then transferred to the general transformer service module **160** for linear data transformation as part of analysis or the decomposable transformer service module **150** for branching or iterative transformations that are part of analysis. The directed computational graph module **155** represents all data as directed graphs where the transformations are nodes and the result messages between transformations edges of the graph. The high volume web crawling module **115** uses multiple server hosted preprogrammed web spiders, which while autonomously configured are deployed within a web scraping framework **115***a* of which SCRAPY™ is an example, to identify and retrieve data of interest from web based sources that are not well tagged by conventional web crawling technology. The multiple dimension time-series data store module **120** may receive streaming data from a large plurality of sensors that may be of several different types. The multiple dimension time-series data store module may also store any time-series data encountered by the system such as but not limited to environmental factors at insured client infrastructure sites, component sensor readings and system logs of all insured client equipment, weather and catastrophic event reports for all regions an insured client occupies, political communiques from regions hosting insured client infrastructure and network service information captures such as, but not limited to news, capital funding opportunities and financial feeds, and sales, market condition and service related customer data. The module is designed to accommodate irregular and high volume surges by dynamically allotting network bandwidth and server processing channels to process the incoming data. Inclusion of programming wrappers for languages examples of which are, but not limited to C++, PERL, PYTHON, and ERLANG™ allows sophisticated programming logic to be added to the default function of the multidimensional time-series database **120** without intimate knowledge of the core programming, greatly extending breadth of function. Data retrieved by the multidimensional time-series database **120** and the high-volume web crawling module **115** may be further analyzed and transformed into task optimized results by the directed computational graph **155** and associated general transformer service **150** and decomposable transformer service **160** modules. Alternately, data from the multidimensional time-series database and high volume web crawling modules may be sent, often with scripted cuing information determining important vertexes **145***a*, to the graph stack service module **145** which, employing standardized protocols for converting streams of information into graph representations of that data, for example, open graph internet technology although the invention is not reliant on any one standard. Through the steps, the graph stack service module **145** represents data in graphical form influenced by any pre-determined scripted modifications **145***a* and stores it in a graph-based data store **145***b* such as GIRAPH™ or a key value pair type data store REDIS™, or RIAK™, among others, all of which are suitable for storing graph-based information.

(21) Results of the transformative analysis process may then be combined with further client directives, additional business rules and practices relevant to the analysis and situational information external to the already available data in the automated planning service module **130** which also runs powerful information theory **130***a* based predictive statistics functions and machine learning algorithms to allow future trends and outcomes to be rapidly forecast based upon the current system derived results and choosing each a plurality of possible business decisions. The using all available data, the automated planning service module **130** may propose business decisions most likely to result is the most favorable business outcome with a usably high level of certainty. Closely related to the automated planning service module in the use of system derived results in conjunction with possible externally supplied additional information in the assistance of end user business decision making, the action outcome simulation module **125** with its discrete

event simulator programming module **125***a* coupled with the end user facing observation and state estimation service **140** which is highly scriptable **140***b* as circumstances require and has a game engine **140***a* to more realistically stage possible outcomes of business decisions under consideration, allows business decision makers to investigate the probable outcomes of choosing one pending course of action over another based upon analysis of the current available data.

(22) A significant proportion of the data that is retrieved and transformed by the business operating system, both in real-world analyses and as predictive simulations that build upon intelligent extrapolations of real-world data, include a geospatial component. The indexed global tile module **170** and its associated geospatial tile manager **170***a* manages externally available, standardized geospatial tiles and may provide other components of the business operating system through programming methods to access and manipulate meta-information associated with geospatial tiles and stored by the system. Ability of the business operating system to manipulate this component over the time frame of an analysis and potentially beyond such that, in addition to other discriminators, the data is also tagged, or indexed, with their coordinates of origin on the globe, allows the system to better integrate and store analysis specific information with all available information within the same geographical region. Such ability makes possible not only another layer of transformative capability but, may greatly augment presentation of data by anchoring to geographic images including satellite imagery and superimposed maps both during presentation of real-world data and simulation runs.

(23) FIG. **2** is a diagram of the web application-based interface for an indexed global tile module **200** as per one embodiment of the invention. User interaction with the indexed global tile module **170** is mediated through a web-based interface **210**. This indexed global tile module web interface **210** may retrieve at least a portion of geospatial data not previously referenced from a plurality of remote, possibly cloud-based sources which include but are not limited to raster tile map sources **260**, interactive map libraries **270** and data and tile rendering services **280** which may resolve business decision data onto a geographical location or region. Other services known to those skilled in the field may exist and be used as needed. Direct interaction between the function of the business operating system **100** and the capabilities of the indexed global tile module may pertain to both real-world analyses and simulations run by the system. The web application therefore keeps galleries for simulations **211**, streaming analyses results **213** and historical analyses results **216** which allows app users to retrieve these records from system data stores which include multidimensional time-series data store **120** for analyses involving series of repeated sensor readings or repeated updates of highly similar data or other similar data series of short to moderate amounts of information and other system associated data stores **218** for analyses of more extensive freeform text or similar data records. Once a simulation or real-world analysis is chosen, users may add simple geospatial information to them such as a unique geo-hash, calculated within the geo-hash server **250** that is closely based upon the geographical coordinates where the data originated that then allows the simulation or real-world analysis to be correlated with other analyses within a pre-determined location radius based upon those geo-hashes which have increasing similarity based upon increased proximity Addition of complex geospatial information such as detailed indexed time corrected geospatial tile images or fully user annotated, labelled time corrected geospatial tile series **260** with map overlays **270** for the length of the analysis, depending on the requirements of the study being carried out. To allow modification of running simulations, the web app must be capable of supplying the key for the correct, desired simulation and therefore keeps a repository of those keys **214**. A plurality of jobs run by the indexed global tile module require both a large amount of resources and of time to complete and may therefore be nm as batch jobs **212**. All output from the manipulation done by the web application may be mediated by the output module. The web application is hosted on a web server **220**. Indexed geospatial tiles are retrieved from at least one tile server which may be local- or cloud-based **230**. Geo-hashes are calculated from specific tiles of interest **250**. Tiles and map overlays may be rasterized for output by a raster server **240**.

(24) FIG. **3** is a diagram of an indexed global tile module **300** as per one embodiment of the invention. A significant amount of the data transformed and simulated by the business operating system has an important geospatial component. The indexed global tile module **170** allows both for the geo-tagged storage of data as retrieved by the system as a whole and for the manipulation and display of data using their geological data to augment the data's usefulness in transformation, for example creating ties between two independently acquired data points to more fully explain a phenomenon or in the display of real-world or simulated results in their correct geospatial context for greatly increased visual comprehension and memorability. The indexed global tile module **170** may consist of a geospatial index information management module which retrieves indexed geospatial tiles from a cloud-based **320** source known to those skilled in the art and may also retrieved available geospatially indexed map overlays **310** for geospatial tiles **320** from a cloud-based source known to those skilled in the art. Tiles and their overlays, once retrieved, represent large amounts of potentially reusable data and are therefore stored for a predetermined amount of time to allow rapid recall during one or more analyses on the system **350**. To be useful it is required that both the transformative modules of the business operating system, such as, but not limited to the directed computational graph module **155**, and the automated planning service module **130**, as well as the action outcome simulation module **125** and observational and state estimation service **140** for display functions be capable of both accessing and manipulating the retrieved tiles and overlays. A geospatial query processor interface serves as a program interface between these system modules and the geospatial index information management module **340** which fulfills the resource requests through specialized direct tile manipulation protocols, which, as a simplistic example, may include “get tile xxx,” “zoom,” “rotate,” “crop,” “shape,” “stitch,” and “highlight” just to name a very few options known to those skilled in the art. During analysis, the geospatial index information management module may control the assignment of geospatial data and the running transformative functions to one or more swimlanes to expedite timely completion and correct storage of the resultant data with associated geotags. The transformed tiles with all associated transformation tagging may be stored for future review **370**. Alternatively, just the geotagged transformation data or geotagged tile views may be stored **370** for future retrieval of the actual tile and review depending on the need and circumstance. There may also be occasion where time-series data from specific geographical locations are stored in the multidimensional time-series data store **120** with geo-tags provided by the geospatial index information management module **340**.

(25) FIG. **4** is a flow diagram illustrating the function of the indexed global tile module **400** as per one embodiment of the invention. Predesignated, indexed geospatial tiles are retrieved from sources known to those skilled in the art **401**. Available map overlay data from one of multiple sources **403** known to those skilled in the art may be retrieved per user design. The geospatial tiles may then be processed in one or more of a plurality of ways according to the design of the running analysis **402**, at which time geo-tagged event or sensor data may be associated with the indexed tile **404** which may be directed programmatically or as a result of user interaction through a web application **210**. Geo-tagging, either interactive **200** or programmatic, may be applied to data at or near real time or applied to transformed data that has been previously archived to augment those data alone for direct presentation or further self-contained analysis, or may be applied to augment the data so as to allow their inclusion in subsequent predictive analyses on the basis of their geographical location of occurrence in larger location based or regionally based analyses with either other archival or real time data, or both. Data relating to tile processing, which may include the tile itself is then stored for later review or analysis **407**. The geo-data, in part, or in its entirety may be used in one or more transformations that are part of a real-world data presentation **405**. The geo-data in part of in its entirety may be used in one or more transformations that are part of a simulation **406**. At least some of the geospatial data may be used in an analyst determined direct visual presentation or may be formatted and transmitted for use in third party solutions **408**.

(26) Hardware Architecture

(27) Generally, the techniques disclosed herein may be implemented on hardware or a combination of software and hardware. For example, they may be implemented in an operating system kernel, in a separate user process, in a library package bound into network applications, on a specially constructed machine, on an application-specific integrated circuit (ASIC), or on a network interface card.

(28) Software/hardware hybrid implementations of at least some of the embodiments disclosed herein may be implemented on a programmable network-resident machine (which should be understood to include intermittently connected network-aware machines) selectively activated or reconfigured by a computer program stored in memory. Such network devices may have multiple network interfaces that may be configured or designed to utilize different types of network communication protocols. A general architecture for some of these machines may be described herein in order to illustrate one or more exemplary means by which a given unit of functionality may be implemented. According to specific embodiments, at least some of the features or functionalities of the various embodiments disclosed herein may be implemented on one or more general-purpose computers associated with one or more networks, such as for example an end-user computer system, a client computer, a network server or other server system, a mobile computing device (e.g., tablet computing device, mobile phone, smartphone, laptop, or other appropriate computing device), a consumer electronic device, a music player, or any other suitable electronic device, router, switch, or other suitable device, or any combination thereof. In at least some embodiments, at least some of the features or functionalities of the various embodiments disclosed herein may be implemented in one or more virtualized computing environments (e.g., network computing clouds, virtual machines hosted on one or more physical computing machines, or other appropriate virtual environments).

(29) Referring now to FIG. **5**, there is shown a block diagram depicting an exemplary computing device **10** suitable for implementing at least a portion of the features or functionalities disclosed herein. Computing device **10** may be, for example, any one of the computing machines listed in the previous paragraph, or indeed any other electronic device capable of executing software- or hardware-based instructions according to one or more programs stored in memory. Computing device **10** may be configured to communicate with a plurality of other computing devices, such as clients or servers, over communications networks such as a wide area network a metropolitan area network, a local area network, a wireless network, the Internet, or any other network, using known protocols for such communication, whether wireless or wired.

(30) In one embodiment, computing device **10** includes one or more central processing units (CPU) **12**, one or more interfaces **15**, and one or more busses **14** (such as a peripheral component interconnect (PCI) bus). When acting under the control of appropriate software or firmware, CPU **12** may be responsible for implementing specific functions associated with the functions of a specifically configured computing device or machine. For example, in at least one embodiment, a computing device **10** may be configured or designed to function as a server system utilizing CPU **12**, local memory **11** and/or remote memory **16**, and interface(s) **15**. In at least one embodiment, CPU **12** may be caused to perform one or more of the different types of functions and/or operations under the control of software modules or components, which for example, may include an operating system and any appropriate applications software, drivers, and the like.

(31) CPU **12** may include one or more processors **13** such as, for example, a processor from one of the Intel, ARM, Qualcomm, and AMD families of microprocessors. In some embodiments, processors **13** may include specially designed hardware such as application-specific integrated circuits (ASICs), electrically erasable programmable read-only memories (EEPROMs), field-programmable gate arrays (FPGAs), and so forth, for controlling operations of computing device **10**. In a specific embodiment, a local memory **11** (such as non-volatile random-access memory (RAM) and/or read-only memory (ROM), including for example one or more levels of cached

memory) may also form part of CPU **12**. However, there are many different ways in which memory may be coupled to system **10**. Memory **11** may be used for a variety of purposes such as, for example, caching and/or storing data, programming instructions, and the like. It should be further appreciated that CPU **12** may be one of a variety of system-on-a-chip (SOC) type hardware that may include additional hardware such as memory or graphics processing chips, such as a Qualcomm SNAPDRAGON™ or Samsung EXYNOS™ CPU as are becoming increasingly common in the art, such as for use in mobile devices or integrated devices.

(32) As used herein, the term “processor” is not limited merely to those integrated circuits referred to in the art as a processor, a mobile processor, or a microprocessor, but broadly refers to a microcontroller, a microcomputer, a programmable logic controller, an application-specific integrated circuit, and any other programmable circuit.

(33) In one embodiment, interfaces **15** are provided as network interface cards (NICs). Generally, NICs control the sending and receiving of data packets over a computer network; other types of interfaces **15** may for example support other peripherals used with computing device **10**. Among the interfaces that may be provided are Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, graphics interfaces, and the like. In addition, various types of interfaces may be provided such as, for example, universal serial bus (USB), Serial, Ethernet, FIREWIRE™, THUNDERBOLT™, PCI, parallel, radio frequency (RF), BLUETOOTH™, near-field communications (e.g., using near-field magnetics), 802.11 (Wi-Fi), frame relay, TCP/IP, ISDN, fast Ethernet interfaces, Gigabit Ethernet interfaces, Serial ATA (SATA) or external SATA (ESATA) interfaces, high-definition multimedia interface (HDMI), digital visual interface (DVI), analog or digital audio interfaces, asynchronous transfer mode (ATM) interfaces, high-speed serial interface (HSSI) interfaces, Point of Sale (POS) interfaces, fiber data distributed interfaces (FDDIs), and the like. Generally, such interfaces **15** may include physical ports appropriate for communication with appropriate media. In some cases, they may also include an independent processor (such as a dedicated audio or video processor, as is common in the art for high-fidelity A/V hardware interfaces) and, in some instances, volatile and/or non-volatile memory (e.g., RAM).

(34) Although the system shown and described above illustrates one specific architecture for a computing device **10** for implementing one or more of the inventions described herein, it is by no means the only device architecture on which at least a portion of the features and techniques described herein may be implemented. For example, architectures having one or any number of processors **13** may be used, and such processors **13** may be present in a single device or distributed among any number of devices. In one embodiment, a single processor **13** handles communications as well as routing computations, while in other embodiments a separate dedicated communications processor may be provided. In various embodiments, different types of features or functionalities may be implemented in a system according to the invention that includes a client device (such as a tablet device or smartphone running client software) and server systems (such as a server system described in more detail below).

(35) Regardless of network device configuration, the system of the present invention may employ one or more memories or memory modules (such as, for example, remote memory block **16** and local memory **11**) configured to store data, program instructions for the general-purpose network operations, or other information relating to the functionality of the embodiments described herein (or any combinations of the above). Program instructions may control execution of or comprise an operating system and/or one or more applications, for example. Memory **16** or memories **11**, **16** may also be configured to store data structures, configuration data, encryption data, historical system operations information, or any other specific or generic non-program information described herein.

(36) Because such information and program instructions may be employed to implement one or more systems or methods described herein, at least some network device embodiments may include nontransitory machine-readable storage media, which, for example, may be configured or designed

to store program instructions, state information, and the like for performing various operations described herein. Examples of such nontransitory machine-readable storage media include, but are not limited to, magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CD-ROM disks; magneto-optical media such as optical disks, and hardware devices that are specially configured to store and perform program instructions, such as read-only memory devices (ROM), flash memory (as is common in mobile devices and integrated systems), solid state drives (SSD) and "hybrid SSD" storage drives that may combine physical components of solid state and hard disk drives in a single hardware device (as are becoming increasingly common in the art with regard to personal computers), memristor memory, random access memory (RAM), and the like. It should be appreciated that such storage means may be integral and non-removable (such as RAM hardware modules that may be soldered onto a motherboard or otherwise integrated into an electronic device), or they may be removable such as swappable flash memory modules (such as "thumb drives" or other removable media designed for rapidly exchanging physical storage devices), "hot-swappable" hard disk drives or solid state drives, removable optical storage discs, or other such removable media, and that such integral and removable storage media may be utilized interchangeably. Examples of program instructions include both object code, such as may be produced by a compiler, machine code, such as may be produced by an assembler or a linker, byte code, such as may be generated by for example a JAVA™ compiler and may be executed using a Java virtual machine or equivalent, or files containing higher level code that may be executed by the computer using an interpreter (for example, scripts written in Python, Perl, Ruby, Groovy, or any other scripting language).

(37) In some embodiments, systems according to the present invention may be implemented on a standalone computing system. Referring now to FIG. **6**, there is shown a block diagram depicting a typical exemplary architecture of one or more embodiments or components thereof on a standalone computing system. Computing device **20** includes processors **21** that may run software that carry out one or more functions or applications of embodiments of the invention, such as for example a client application **24**. Processors **21** may carry out computing instructions under control of an operating system **22** such as, for example, a version of Microsoft's WINDOWS™ operating system, Apple's Mac OS/X or iOS operating systems, some variety of the Linux operating system, Google's ANDROID™ operating system, or the like. In many cases, one or more shared services **23** may be operable in system **20**, and may be useful for providing common services to client applications **24**. Services **23** may for example be WINDOWS™ services, user-space common services in a Linux environment, or any other type of common service architecture used with operating system **21**. Input devices **28** may be of any type suitable for receiving user input, including for example a keyboard, touchscreen, microphone (for example, for voice input), mouse, touchpad, trackball, or any combination thereof. Output devices **27** may be of any type suitable for providing output to one or more users, whether remote or local to system **20**, and may include for example one or more screens for visual output, speakers, printers, or any combination thereof. Memory **25** may be random-access memory having any structure and architecture known in the art, for use by processors **21**, for example to run software. Storage devices **26** may be any magnetic, optical, mechanical, memristor, or electrical storage device for storage of data in digital form (such as those described above). Examples of storage devices **26** include flash memory, magnetic hard drive, CD-ROM, and/or the like.

(38) In some embodiments, systems of the present invention may be implemented on a distributed computing network, such as one having any number of clients and/or servers. Referring now to FIG. **7**, there is shown a block diagram depicting an exemplary architecture **30** for implementing at least a portion of a system according to an embodiment of the invention on a distributed computing network. According to the embodiment, any number of clients **33** may be provided. Each client **33** may run software for implementing client-side portions of the present invention; clients may comprise a system **20** such as that illustrated above. In addition, any number of servers **32** may be

provided for handling requests received from one or more clients **33**. Clients **33** and servers **32** may communicate with one another via one or more electronic networks **31**, which may be in various embodiments any of the Internet, a wide area network, a mobile telephony network (such as CDMA or GSM cellular networks), a wireless network (such as Wi-Fi, WiMAX, LTE, and so forth), or a local area network (or indeed any network topology known in the art; the invention does not prefer any one network topology over any other). Networks **31** may be implemented using any known network protocols, including for example wired and/or wireless protocols.

(39) In addition, in some embodiments, servers **32** may call external services **37** when needed to obtain additional information, or to refer to additional data concerning a particular call. Communications with external services **37** may take place, for example, via one or more networks **31**. In various embodiments, external services **37** may comprise web-enabled services or functionality related to or installed on the hardware device itself. For example, in an embodiment where client applications **24** are implemented on a smartphone or other electronic device, client applications **24** may obtain information stored in a server system **32** in the cloud or on an external service **37** deployed on one or more of a particular enterprise's or user's premises.

(40) In some embodiments of the invention, clients **33** or servers **32** (or both) may make use of one or more specialized services or appliances that may be deployed locally or remotely across one or more networks **31**. For example, one or more databases **34** may be used or referred to by one or more embodiments of the invention. It should be understood by one having ordinary skill in the art that databases **34** may be arranged in a wide variety of architectures and using a wide variety of data access and manipulation means. For example, in various embodiments one or more databases **34** may comprise a relational database system using a structured query language (SQL), while others may comprise an alternative data storage technology such as those referred to in the art as “NoSQL” (for example, Hadoop Cassandra, Google BigTable, and so forth). In some embodiments, variant database architectures such as column-oriented databases, in-memory databases, clustered databases, distributed databases, or even flat file data repositories may be used according to the invention. It will be appreciated by one having ordinary skill in the art that any combination of known or future database technologies may be used as appropriate, unless a specific database technology or a specific arrangement of components is specified for a particular embodiment herein. Moreover, it should be appreciated that the term “database” as used herein may refer to a physical database machine, a cluster of machines acting as a single database system, or a logical database within an overall database management system. Unless a specific meaning is specified for a given use of the term “database”, it should be construed to mean any of these senses of the word, all of which are understood as a plain meaning of the term “database” by those having ordinary skill in the art.

(41) Similarly, most embodiments of the invention may make use of one or more security systems **36** and configuration systems **35**. Security and configuration management are common information technology (IT) and web functions, and some amount of each are generally associated with any IT or web systems. It should be understood by one having ordinary skill in the art that any configuration or security subsystems known in the art now or in the future may be used in conjunction with embodiments of the invention without limitation, unless a specific security **36** or configuration system **35** or approach is specifically required by the description of any specific embodiment.

(42) FIG. **8** shows an exemplary overview of a computer system **40** as may be used in any of the various locations throughout the system. It is exemplary of any computer that may execute code to process data. Various modifications and changes may be made to computer system **40** without departing from the broader scope of the system and method disclosed herein. Central processor unit (CPU) **41** is connected to bus **42**, to which bus is also connected memory **43**, nonvolatile memory **44**, display **47**, input/output (I/O) unit **48**, and network interface card (NIC) **53**. I/O unit **48** may, typically, be connected to keyboard **49**, pointing device **50**, hard disk **52**, and real-time clock **51**.

NIC **53** connects to network **54**, which may be the Internet or a local network, which local network may or may not have connections to the Internet. Also shown as part of system **40** is power supply unit **45** connected, in this example, to a main alternating current (AC) supply **46**. Not shown are batteries that could be present, and many other devices and modifications that are well known but are not applicable to the specific novel functions of the current system and method disclosed herein. It should be appreciated that some or all components illustrated may be combined, such as in various integrated applications, for example Qualcomm or Samsung system-on-a-chip (SOC) devices, or whenever it may be appropriate to combine multiple capabilities or functions into a single hardware device (for instance, in mobile devices such as smartphones, video game consoles, in-vehicle computer systems such as navigation or multimedia systems in automobiles, or other integrated hardware devices).

(43) The geospatial labelling platform described here is built upon highly programmable computer software architecture that may serve as the basis of a plurality of specific use systems. For example the architecture and base programming described here **100** being employed as an trading decision platform **200** is the same computer architecture described in ¶032 and ¶033 of co-pending application Ser. No. 15/237,625 and specifically used as a cyber-attack detection mitigation and remediation platform in ¶035 through ¶037 of co-pending application Ser. No. 15/237,625 all of which may benefit from the inclusion of geographical location data in way evident to those skilled in the art. The same base architecture and programming, presented here and previously and designed to be readily augmented by application specific data stores and programming may take on the capabilities or personalities of a plurality of highly advanced platforms in a plurality of fields both business and scientific where large volumes of data, at least a portion of which may enter the system in bursts or at irregular intervals is present and data which may need normalization and transformation as well as correlation of possibly hard to discern commonalities. Again, the ability by the system to retrieve and include geospatial data even in the most simplistic addition of geo-hashes may greatly benefit achievement of progress in a number of the mentioned areas. The personality instilled platform may also be used in these fields to perform reliable analytics and run reliable simulations on the existing data to allow operators to intelligently determine next direction to implement (and which next direction potentially not to implement) potentially saving both time, money and resources. In summary, the business operating system disclosed here, now with the addition of a mechanism to process geographical data and add it to both currently accruing and archival data and in co-pending applications may be imagined more as a set of software engineered stations in a highly and readily modifiable virtual production line than as only a cyber-attack detection, mitigation and remediation system or as only and trading decision platform as it is both and may be more.

(44) In various embodiments, functionality for implementing systems or methods of the present invention may be distributed among any number of client and/or server components. For example, various software modules may be implemented for performing various functions in connection with the present invention, and such modules may be variously implemented to run on server and/or client.

(45) The skilled person will be aware of a range of possible modifications of the various embodiments described above. Accordingly, the present invention is defined by the claims and their equivalents.

## Claims

1. A system for highly scalable parallel world simulations, comprising: a plurality of computing devices each comprising at least a processor, a memory, and a network interface; wherein a plurality of programming instructions stored in one or more of the memories and operating on one or more of the processors of the plurality of computing devices causes the plurality of computing

devices to: retrieve a plurality of geospatial image tiles corresponding to a geographic region from a plurality of sources; retrieve a plurality of geotagged data corresponding to the geographic region; calculate a geohash for each piece of retrieved geotagged data, wherein a geohash is an encoded geographic location comprising a short string of letters and digits; overlay the geohash on the corresponding geospatial image tile containing the geographic coordinates of the geohash; assign each geographic region and the corresponding geotagged data to a unique process execution swimlane; execute simulations within each unique process execution swimlane using its respective geotagged data; generate updated geospatial image tile data based on results of the simulations; transmit the updated geospatial image tile data to a user via a network; and send map overlay data for at least one of the transmitted updated geospatial image tiles to the user via the network; wherein the geohashes for each pair of geographic regions have a degree of string similarity that is inversely proportional to the geographic distance between the respective geographic regions, such that geohashes for proximate geographic regions share longer common prefixes than geohashes for distant geographic regions.

2. The system of claim 1, wherein the geographic region is a geographical location made up of at least one geographical point.

3. The system of claim 1, wherein the geographic region is a geographical region made up of a plurality of geographical locations.

4. The system of claim 1, wherein post-simulation data is stored and retrieved for further analysis.

5. The system of claim 1, wherein the plurality of geotagged data comprises streaming data from sensors.

6. The system of claim 1, wherein executing simulations comprises running parallel world simulations on aggregated geospatial image tiles and corresponding geotagged data for a geographic region to predict real-world outcomes.

7. The system of claim 1, wherein: the plurality of geotagged data comprises sensor data from multiple real-world locations within the geographic region; and the simulations generate predicted future sensor data values for those locations.

8. A method for highly scalable parallel world simulations, comprising the steps of: retrieving a plurality of geospatial image tiles corresponding to a geographic region from a plurality of sources; retrieving a plurality of geotagged data corresponding to the geographic region; calculating a geohash for each piece of retrieved geotagged data, wherein a geohash is an encoded geographic location comprising a short string of letters and digits; and overlaying the geohash on the corresponding geospatial image tile containing the geographic coordinates of the geohash; assigning each geographic region and the corresponding geotagged data to a unique process execution swimlane; executing simulations for within each unique process execution swimlane using its respective geotagged data; generating updated geospatial image tile data based on results of the simulations; transmitting the updated geospatial image tile data to a user via a network; and sending map overlay data for at least one of the transmitted updated geospatial image tiles to the user via the network; wherein the geohashes for each pair of geographic regions have a degree of string similarity that is inversely proportional to the geographic distance between the respective geographic regions, such that geohashes for proximate geographic regions share longer common prefixes than geohashes for distant geographic regions.

9. The method of claim 8, wherein the geographic region is a geographical location made up of at least one geographical point.

10. The method of claim 8, wherein the geographic region is a geographical region made up of a plurality of geographical locations.

11. The system of claim 8, wherein post-simulation data is stored and retrieved for further analysis.

12. The system of claim 8, wherein the plurality of geotagged data comprises streaming data from sensors.

13. One or more non-transitory computer-storage media having computer-executable instructions

embodied thereon that, when executed by one or more processors of a computing system employing a similarity subsystem, cause the computing system to perform the method of claim 8.
14. The method of claim 8, wherein executing simulations comprises running parallel world simulations on aggregated geospatial image tiles and corresponding geotagged data for a geographic region to predict real-world outcomes.
15. The method of claim 8, wherein: the plurality of geotagged data comprises sensor data from multiple real-world locations within the geographic region; and the simulations generate predicted future sensor data values for those locations.

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